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SUPERSEDING
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16 May 1980

MILITARY SPECIFICATION

CANNON, 155MM HOWITZER: M199

This specification is approved for use within the Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification supports the acquisition of the Cannon, 155MM Howitzer: M199 consisting of a barrel assembly (tube and muzzle brake) a breech assembly, a lanyard assembly, and a thermal warning device. This specification includes the minimum essential Engineering and Packaging Requirements and the necessary Quality Assurance Provisions to determine that these requirements have been met.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified in the solicitation, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

Military

MIL-C-13931 - Cannon, General Specification for

MIL-I-45607 - Inspection Equipment, Acquisition, Maintenance and Disposition of

MIL-H-45984 - Howitzer, Medium, Towed, 155MM: M198

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Armament Research, Development and Engineering Center, US Army Armament, Munitions and Chemical Command, ATTN: SMCAR-CCB-SAS, Watervliet, NY 12189-4050 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1025

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STANDARDS

Military

MIL-STD-129 - Marking for Shipment and Storage

MIL-STD-1189 - Standard Department of Defense Bar Code Symbology

MIL-STD-1235 - Single and Multi-Level Continuous Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-2073-1 - DOD Materiel Procedures for Development and Application of Packaging Requirements

2.1.2 Government drawings. The following Government drawing form a part of this specification to the extent specified herein. Unless otherwise specified in the solicitation, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

US Army Armament Research, Development and Engineering Center

EPL 11578880 - Engineering Parts List, Cannon, 155MM Howitzer: M199

SPI P11578880 - Special Packing Instruction, Cannon, 155MM Howitzer: M199

DL 11578880 - List of Inspection Equipment for Cannon, 155MM Howitzer: M199

Master LQAP 11578880 - Master List of Quality Assurance Provisions for Cannon, 155MM Howitzer: M199

QAP-APPENDIX WVA - General Quality Assurance Provisions

(Copies of specifications, standards, handbooks, drawings, publications and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the Procuring Contracting Officer.)

2.2 Other publications. The following documents (s) form a part of this specification to the extent specified herein. Unless otherwise specified in the solicitation, the issues of th e documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified in the solicitation, the issures of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

American Society of Testing and Materials (ASTM)

ASTM D3951 - Commercial Packaging, Practice for

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. The applicable contract contains an order of precedence clause wherein the specifications are listed last. With respect to this specification, the order of precedence is drawing requirements, followed by the text of this specification, Quality Assurance Provisions (QAPs), then the referenced specifications and standards. Nothing in this specification shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General requirements. Cannon and component parts shall be in accordance with the applicable contract, the applicable drawings, Engineering Parts List 11578880, the requirements of this specification, the provisions of MIL_C_13931, and all referenced documents. Any conflicts are to be resolved by means of the order of precedence (see 2.3).

3.2 First article. The contractor shall submit a first article unless it is specifically waived in the contract (see 4.4 and 6.2). No first article requirements shall be waived without review and approval by the Procuring Contracting Officer (see 6.4).

3.3 Reliability, Availability, Maintainability and Durability (RAM-D). The RAM-D (see 50.1 - Appendix B) requirements of the M198 Howitzer system for the Operational Mode Summary/Mission Profile (OMS-MP) are stated in MIL-H-45984. Since the availability arid maintainability requirements are specified for the M198 Howitzer system only, these attributes cannot be apportioned for the cannon; but, the reliability and durability requirements are specified for the cannon in Appendix B (see 30).

3.4 Functional characteristics. The cannon shall perform the firing, pressure sustaining, and projectile launching functions of the 155MM M198 Towed Howitzer System. The breech mechanism, firing mechanism and thermal warning device shall function smoothly without interference, erratic movement or malfunction.

3.4.1 Breech mechanism. To open the breech, a clockwise movement of the lower mechanism. handle shall unlatch the release lever. Continued clockwise rotation of the breechblock assembly to the point where the upper handle caroming surface contacts the breech ring cam shall disengage the sectored threads. At this point, the block lock shall engage the cam plate slot to prevent any rotation of the breechblock while it is free to swing.

The disengaged breechblock assembly shall swing about the carrier hinge pin which is located at 45° from the vertical centerline. The opening swing shall be controlled by a rearward force on the handles assisted by gravity. The swing shall be limited by contact between the carrier stop surface and the breech ring.

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To close the breech, an upward force on the handles, assisted by the spring force of the counterbalance assembly, shall swing the breechblock assembly up to the point where the block lock disengages and the upper handle caroming surface contacts the breech ring cam to initiate counter-clockwise rotation of the breechblock and engagement of the sectored threads. Continued counterclockwise rotation to the point where the cam plate inner stop surface contacts the housing assembly shall bring the breech block assembly into the fully closed position. In this position, the handle release lever shall engage the locking dowel.

3.4.2 Firing mechanism block assembly. As the breechblock assembly is rotated clockwise to open, the firing mechanism block assembly shall be automatially caromed horizontally to the left. During this movement, the extractor arms shall be actuated to forcibly eject the primer case from the primer chamber. When the breechblock assembly is in the disengaged position, the firing mechanism block assembly shall be removable by sliding the assembly horizontally to the right while pulling outward on the cam follower.

As the breechblock assembly is rotated counterclockwise to close, the firing mechanism block assembly shall be automatically caromed horizontally to the right from the extract position to reach the primer load position as the breechblock assembly reaches the fully closed position. When the breech is closed, the firing mechanism block assembly shall be capable of being moved manually between the load, the extract, and the firing positions.

3.4.3 Obturator group. The obrurator group functioning requirements shall conform to MIL-C-393 ; however, the bearing requirements of obturator parts shall apply when a force of 11,000 - 12,000 pounds is applied to the spindle in a direction tending to compress the pad.

3.4.4 Firing mechanism. When assembled in the cannon, the firing mechanism assembly shall cock and fire the primer as its lever is pulled by the attached lanyard. When the lanyard is released, the mechanism shall position itself for the next cycle.

3.4.5 Primer headspace. Primer headspace, the distance between the front face of the firing mechanism block assembly and the rear face of the obturator spindle, shall not be less than 0.062 inch nor more than 0.074 inch.

3.4.6 Firing pin protrusion and retraction. The firing pin protrusion measured from the front face of the firing mechanism block shall be not less than 0.102 inch nor more than 0.119 inch in the fired position. Firing pin retraction behind the front face of firing mechanism block shall be not less than 0.004 inch.

3.4.7 Thermal warning device. The requirements for the thermaal warning device are specified in Appendix A (see 30.4).

3.5 Performance.

3.5.1 High pressure resistance. The cannon, spare tubes and muzzle brakes shall be capable of withstanding a chamber pressure of 59,000 ± 2,300 psi at 70° + 3°F.

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3.5.2 Material soundness. After completion of performance testing, all parts shall be free from cracks and fractures and shall comply with soundness requirements specified on the applicable drawings.

3.5.3 Bore enlargement. Before diameters across lands, after firing no more than 15 rounds, shall not exceed the before firing diameters by more than the amounts shown in Table I.

TABLE I. Bore enlargement.

Distance forward of commencement of rifling	0.10 inch	1 inch	1 cal	2 cals to muzzle
Amount (in inches)	0.008	0.008	0.006	0.004

NOTE: Interpolate to determine the allowable enlargement between the distances listed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein,

perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to the prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of Sections 3 and 5. The inspections set forth in this specifications shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in this specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Governemnt to acceptance of defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. FA-IPT inspection (see 4.5).

c. RAM-D inspection (see 4.6).

d. Quality conformance inspection (4.7).

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4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in the applicable test document, applicable paragraph(s) in the specification, or in the applicable drawing.

4.4 First article inspection. Unless otherwise specified in the contract (see 6.2.1), first article items shall be submitted for inspection in accordance with QAP-APPENDIX-WVA and applicable item Quality Assurance Provisions (QAPs). First article items shall be representative of the production processes used during quantity production. QAPs are listed in Master LQAP 11578880. First article quantities are specified in the applicable item QAPs.

4.5 First Article - Initial Production Testing (FA-IPT). The requirements, tests, and test methods for FA-IPT are contained in Appendix A.

4.6 Reliability, Availability, Maaintainability, and Durability (RAM-D). The requirements, tests, and test methods for RAM-D are contained in Appendix B.

4.7 Quality conformance inspection.

4.7.1 Inspection provisions. Cannon and parts thereof shall be inspected for conformance to the requirements of this specification, to MIL-C-13931, QAPs listed in Master LQAP 11578880 and QAP-APPENDIX-WVA.

4.7.2 Inspection approval stamp. The application of the inspection approval stamp shall be as specified in MIL-C-13931.

4.7.3 Inspection equipment.

4.7.3.1 Acquisition, maintenance and disposition. Unless otherwise specified (see 6.2.1) responsibility for acquisition, maintenance, and disposition of inspection equipment listed in List of Inspection Equipment DL 11578880, shall be in accordance with MIL-I-45607.

4.7.3.2 Accuracy of commercial inspection equipment. When commercial and modified commercial inspection equipment is used, it shall be capable of repetitive measurements to an accuracy of 10 percent of the total tolerance of the characteristic being inspected.

4.7.3.2.1 Accuracy of proof pressure measuring devices. The devices for measuring proof firing pressures shall have the following accuracy:

Device	Maximum error of measurement (%)
copper crusher gages	± 3 of reading
pressure transducers	± 2 of reading

4.7.4 Examinations.

4.7.4.1 Primer headspace. The primer headspace shall be gaged on each cannon with gage prescribed in applicable item QAP, or with Government approved

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equivalent gage. Failure to comply with the requirements of 3.4.5 shall be cause for rejection.

4.7.4.2 Firing pin protrusion and retraction. Firing pin protrusion and retraction shall be gaged on each cannon with the gages prescribed in the applicable item QAP, or with Government approved equivalent gages. Failure to comply with the requirements of 3.4.6 shall be cause for rejection.

4.7.4.3 Material soundness.

4.7.4.3.1 Tube. The outer and inner surfaces of each tube shall be magnetic particle inspected in accordance with applicable drawings, QAPs, and specifications. A borescope shall be used during inspection of the tube bore. Failure to comply with the requirements of 3.5.2 shall be cause for rejection.

4.7.4.3.2 Breech and muzzle brake. All surfaces of each breechblock, breech ring, and muzzle brake shall be magnetic particle inspected in accordance with applicable drawings, QAPs, and specifications. Failure to comply with the requirements of 3.5.2 shall be cause for rejection.

4.7.4.4 Packaging. The inspections to determine compliance with the packaging requirements of Section 5, for the level of protection required in the contaract, shall be as specified in MIL-STD-2073-1.

4.7.5 Functional inspection.

4.7.5.1 Breech mechanism and firing mechanism. This inspection shall be done either manually or through the use of a functional tester. A minimum of five complete cycles of breech operation shall be performed on each breech mechanism and firing mechanism. Each time the breechblock assembly reaches the fully closed position, the firing mechanism block assembly shall be cycled between the load, the extract, and the firing positions. Each time the firing mechanism block assembly is cycled into the firing position, the lever shall be pulled and released. Failure to comply with the requirements of 3.4.1, 3.4.2, and 3.4.4 shall be cause for rejection.

4.7.5.2 Obturator group. The obturator group surfaces and the bearing of the obturator parts shall be inspected by using applicable inspection equipment prescribed in applicable item QAPs, or by using Government approved equivalent inspection equipment. Failure to comply with the requirements of 3.4.3 shall be cause for rejection.

4.7.6 Tests and test methods.

4.7.6.1 Proof acceptance. Cannon shall be accepted through proof firing, simulation testing, or both. Failure to comply with 3.5 shall be cause for rejection.

4.7.6.1.1 Breech mechanisms. Each breech mechanism, including all breech mechanisms which are a part of cannon to be proof fired, shall be simulation tested using the method of 4.7.6.2.1.

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4.7.6.1.2 Cannon and spare tubes. Unless otherwise specified by the Procuring Contracting Officer, cannon and spare tubes shall be sampled for proof firing (see 4.7.6.2.2), in accordance with MIL-STD-1235, using plan CSP-T at an AQL of 1.0. The appropriate sampling parameters, "i" and "f", shall be designated by the Procuring Contracting Officer in the contract (see 6.2). Cannon which are or are not proof fired as a result of the sampling procedure shall have had their breech mechanisms simulation tested and accepted (see 4.7.6.2.1).

4.7.6.1.3 Successive production buys. Successive production buys shall not require "i" quantities. However, a new "i" quantity shall be used if any of the following conditions apply:

a. The time lapse between two successive buys from the same contractor is one year-or more.

b. A new contractor is utilized.

c. Significant change is made in a manufacturing process or material.

d. A design change is made.

4.7.6.1.4 Weapon record data. Each cannon and spare tube shall be furnished with applicable data on DA Form 2408-4 (Weapon Record Data), as specified by the Procuring Contracting Officer.

4.7.6.2 Proof acceptance test methods.

4.7.6.2.1 Breech mechanisms. Breech mechanisms shall be simulation tested at a Government specified weapon test facility in accordance with the following procedure:

a. The breech mechanism shall be tested once by means of hydraulic pressure testing equipment at a pressure of 59,000 + 2,300 psi. Test pressure shall be immediately released upon its attainment. The pressure-time graph shall be subject to approval by the Procuring Contracting Officer.

(1) When difficulty is encountered in attaining test pressure, the number of attempts in which over 10 percent of the test pressure is attained, shall not exceed five.

(2) If the cause of difficulty cannot be identified and corrected within five pressure cycles, the breech mechanism shall be subjected to proof firing as part of the cannon (see 4.7.6.2.2). This fact shall be noted on inspection records identifiable with the specific breech mechanism.

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b. The peak pressure shall be recorded for each test and added to inspection records identifiable with the specific breech mechanism undergoing test.

c. After the pressure test:

(1) The breech mechanism shall be examined for looseness or interference of fit.

(2) The breech mechanism shall be disassembled and examined for soundness of material. All components shall be visually examined to assure freedom from cracks and fractures. The breech ring and breechblock shall be magnetic particle inspected (see 4.7.4.3.2).

(3) The breech mechanism shall be cleaned and reassembled.

(4) The breech mechanism and firing mechanism shall be functioned (see 4.7.5.1). In addition, two dummy promers (max and Min metal) shall be used to check headspace, firing pin protrusion and retraction, and ejection.

4.7.6.2.2 Cannon and spare tubes. Cannon and spare tubes shall be proof fired at a Government specified proving ground in accordance with the following procedure:

a. Cannon and spare tubes shall be proof fired once at 59,000 ± 2,300 Psi.

b. A facility mount may be used if necessary.

c. Chamber pressure shall be measured and recorded.

d. After proof firing:

(1) The bore of each tube shall remeasured for bore enlargement using an air gage, an electronic gage, or other approved gage. The measurements shall be recorded. Bore enlargement shall meet the requirements of 3.5.3 and Table I.

(2) The tube shall be magnetic particle inspected and borescoped (see 4.7.4.3.1).

(3) The muzzle brake shall be magnetic particle inspected (see 4.7.4.3.2).

(4) The breech mechanism shall be examined for looseness or interference of fit.

(5) The breech mechanism shall be disassembled and examined for soundness of material. All components shall be visually examined to assure freedom from cracks and fractures. The breech ring and breechblock shall be magnetic particle inspected (see 4.7.4.3.2).

(6) The breech mechanism shall be cleaned and reassembled. It shall then be functioned in accordance with 4.7.5.1.

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e. Cannon or spare tubes found to be defective at the proving ground shall be reported to the Procuring Contracting Officer; the Product Assurance Directorate, Watervliet Arsenal, Watervliet, NY 12189-4050; and Benet Laboratories, Watervliet Arsenal, Watervliet, NY 12189-4050; within one working day. The test agency shall forward a Test Incident Report (TIR) of the incident within three working days.

5. PACKAGING

5.1 Cannon - levels A, B and C. Cleaning, drying, preservation, unit packaging and packing of cannon shall be in accordance with the requirements of Special Packaging Instruction (SPI) P11578880 and MIL-STD-2073-1 for the level of protection specified in the contract (see 6.2).

5.2 Repair parts - levels A, B and C. Cleaning, drying, preservation, unit packaging and packing of repair parts, shall be in accordance with the applicable repair part SPI and MIL-STD-2073-1 for the level of protection specified in the contract (see 6.2).

5.2.1 Level X - industrial packaging. Cleaning, drying, preservation, unit packaging and packing or repair parts shall be in accordance with ASTM D3951 when industrial packaging is specified in the contract (see 6.2).

5.2.1.1 Quality in unit package (QUP). When industrial packaging of repiar parts is specified in the contract, the QUP shall be the same as specified in the contract (see 6.2).

5.3 Marking of packages.

5.3.1 Levels A, B and C. Marking of the unit package, intermediate packages, and exterior shipping containers shall be in accordance with MIL-STD-129. When specified (see 6.2), bar code marking shall be in accordance with MIL-STD-129 and MIL-STD-1189.

5.3.2 Level X - industrial. Marking of the unit package, intermediate packages, and exterior shipping containers shall be in accordance with ASTM D3951. When specified (see 6.2), bar code marking shall be in accordance with MIL-STD-129 and MIL-STD-1189.

6. NOTES

6.1 Intended use. The cannon specified herein is intended for use with the Howitzer, Medium, Towed, 155MM: M198.

6.2 Ordering data.

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6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Requirements for and the number of cannon, components thereof, or both to be submitted for first article by the contractor (see 3.2),

c. The examinations and tests to be performed by the contractor and the examinations and tests to be performed by the Government.

d. All inspection and test data that is required.

e. At whose expense a retest may be performed.

f. Serial numbers for the cannon and repair tube assemblies.

g. Inspection equipment, responsibility for acquisition, maintenance, and disposition thereof, if other than as specified (see 4.7.3).

h. Availability of inspection equipment from the Government.

i. Extent of supplier's responsibility for Government furnished and for supplier-required final inspection equipment.

j. Applicable acceptance test procudcers

k. When a proof firing sampling plan is authorized, the proof firing sampling plan parameters: initial consecutive quantity (i), and sampling frequency (f) (see 4.7.6.1.2).

l. Under what conditions simulated firing may be performed, the test equipment available, and the procedures to be followed.

m. The levels of preservation, packaging, and packing required for cannon assemblies (see 5.1) and repair parts (see 5.2)

n. When applicable, bar code marking requirements (see 5.3).

o. When warranted, the application of MIL-Q-9858 or MIL-I-45208, *as* appropriate.

p. Unless otherwise specified, the application of MIL-I-45607 and MIL-STD-45662.

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6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirement List (CDRL), the data requirements identified below will be developed as specified by an-approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DOD Far Supplement, Part 27, Sub-Part 27.410-6 are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs.

Paragraph No.	Date Requirement Title	Applicable DID No.
4.4	First Article Inspection Report	DI-T-4902

(Copies of Data Item Descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the Procuring Contracting Officer.)

6.4 Waivers and deviations. The Procuring Contracting Officer shall coordinate all requests for waiver or deviation to this specification with Benet Laboratories and the Product Assurance Directorate at Watervliet Arsenal.

6.5 Subject term (key word) listing.

Cannon, 155MM Howitzer MISC
Howitzer, Medium, Towed, 155MM: M198, for use with

6.6 Changes from previous issue. Asterisks (or verticle lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:

Army - AR

Preparing activity:

Army - AR

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APPENDIX A

FIRST ARTICLE - INITIAL PRODUCTION TESTING REQUREMENTS
AND INSPECTION METHODS FOR CANNON, 155MM: M199

10. SCOPE

10.1 Scope. This Appendix details all requirements and inspection methonds necessary for First Article - Initial Production Testing (FA-IPT). This Appendix A is a mandatory part of the specification. The information contained herein is intended for compliance.

20. APPLICABLE DOCUMENTS. This section is not applicable to this Appendix A.

30. REQUIREMENTS

30.1 FA-IPT testing. FA-IPT tests shall be conducted on cannon or critical assemblies or parts thereof when:

a. A new contractor is utilized.

b. A time lapse of one year has occurred since the item was last produced by the contractor.

c. Significant changes have been made in a manufacturing process or material.

d. A design change is made.

If the cannon or critical assemblies or parts thereof fail FA-IPT testing, then failure analysis shall be invoked (see 40.4),

30.2 Performance. The requirements for performance are contained in 3.5.1 3.5.2, and 3.5.3 of MIL-C-45975.

30.3 Climatic. The cannon will be employed in all types of areas, terrain and climates in which the supported units may operate. This includes operation in geographical areas which include temperature extremes from +125°F to -50°F.

30.4 Thermal warning device. The thermal warning device asembly fastened to the top of the breech ring, shall indicate the approximate temperature of the tube powder" chamber and warn the user of a potentially dangerous overheated tube condition. The indicator shall rotate freely and smoothly about its pivot over a range of 0°F to 500°F when the sensing bulb is subjected to temperature changes. Within three minutes after firing 12 zone 8 M203 rounds at four rounds per minute followed by eight rounds of the same type at two rounds per minute, the temperature registered on the device shall increase at least 75°F above its initial reading.

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30.5 Reliability, Availability, Maintainability and Durability (RAM-D). The requirements for RAM-D are contained in Appendix B of MIL-C-45975. When specified in the contract (see 6.2.1), RAM-D tests shall be conducted after successful completion of FA-IPT (Appendix A of MIL-C-45975).

40. INSPECTION METHODS

40.1 First Article - Initial Production Testing (FA-IPT). When specified in the contract (see 6.2.1), first article cannon shall be subjected to FA-IPT by a Government activity to assure compliance with the requirements of this specification and the contract.

40.1.1 Sample. Four first article cannon shall be randomly selected from the first month's production. First article cannon and parts thereof shall be representative of the production processes used during quantity production. When specified, these cannon shall be used for RAM-D testing.

40.1.2 Inspection provisions. First article cannon shall be subjected to normal inspection per applicable item QAP's, quality conformance inspection (see 4.7); including proof acceptance (see 4.7.6.1 and 4.7.6.2), through proof firing and simulation testing.

40.1.2.1 Physical configuration audit. One cannon selected from the first article sample shall undergo 100 percent inspection (no sampling) of all characteristics in addition to the normal inspection cited in 40.1.2.

40.1.2.2 Proof acceptance. First article cannon shall have succesfully passed proof acceptance (see 4.7.6.1 and 4.7.6.2), through proof firing and simulation testing, prior to commencement of FA-IPT.

40.1.3 Tests. Unless otherwise specified herein or in the contract, first article cannon tests listed in this specification shall be performed in a fully operational Howitzer 155MM: M198, and shall not normally be integrated with other tests. Intergration with Operational Testing by the user is allowable provided all components of the system conform to technical requirements. All tests shall be conducted by the Government or any agency specified by the Government.

40.2 TESTS

40.2.1 Climatic test. A climatic test shall be conducted on four cannon using the test method of 40.3.1. If the requirements of 30.0 are not complied with, acceptance shall be deferred and 40.4 shall apply.

40.2.2 Thermal warning device test. A thermal warning device test shall be conducted on four cannon using the test method of 40.3.2. If the requirements of 30.4 are not complied with, acceptance shall be deferred and 40.4 shall apply.

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40.2.3 RAM-D tests. The tests to verify the RAM-D requirements are contained in Appendix B of MIL-C-45975.

40.3 TEST METHODS

40.3.1 Climatic test. A climatic test shall be conducted on four cannon with both the cannon and ammunition conditioned to the temperature specified in Table II. Ammunition used in this test sshall be inert loaded M483A1 projectiles with dummy PD M51A5 fuzes, M82 primers, and zone 8s M203 charge. Magnetic particle (see 4.7.4.3.l and4.7.4.3.2) and borescope (see 4.7.4.3.1) examinations shall be conducted on each cannon after firing in accordance with Table II. Facilities components may be used if necessary. If the requirements of 30.3 are not complied with, acceptance shall be deferred and paragraph 40.4 applies.

TABLE II. Climatic test schedule.

No. rounds	Temperature of cannon and ammunition
5	+125°F + 5°F
5	-25°F + 5°F
5	-50°F + 5°F

NOTE ; DO NOT use live M483A1's as this could result in a catastrophic failure.

40.3.2 Thermal warning device test. Four cannon shall be fired a t 800 + 10 mils elevation in accordance with the rate of firing schedule outlined in Table III. Ammunition used in this test shall be inert loaded M483A1 projectiles with dummy PD M51A5 fuzes, M82 primers, and zone 8 M203 charge, at 70°F ± 3°F. The temperature indicated on the thermal warning device shall be recorded immediately before and within three minutes after the test. If the requirements of 30.4 are not complied with, acceptance shall be deferred and paragraph 40.4 applies.

TABLE III. Rate of fire.

Sequence	Rate of fire	No. of rounds
1	4 rds/min	12
2	2 rds/min	8

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40.4 Failure analysis. The contractor shall conduct a failure analysis by performing a dimensional, physical and visual examination of the components which are suspected to have caused the failure. Results shall be submitted to the Government for a failure cause determination. If it is determined that failure occurred because of deviation from specified dimensions or physical properties or both, the contractor shall accomplish, as applicable, the following actions:

a. Evaluate and correct the applicable production processes and procedures to prevent recurrence of the same defect in future production.

b. Examine completely assembled cannon and partially assembled cannon and components to ensure that material containing the same defect is purged from the inventory and not presented to the Government for acceptance.

c. Submit the results of the failure analysis and the corrective actions taken to the Government for evaluation and approval of retest or continuation of test.

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APPENDIX B

RELIABILITY, AVAILABILITY, MAINTAINABILITY, DURABILITY REQUIREMENTS AND RELIABILITY SCORING CRITERIA FOR CANNON, 155MM: M199

10. SCOPE

10.1 Scope. This Appendix details all requirements and inspetion methods necessary for Reliability, Availability, Maintainability and Durability. This Appendix B also provides Reliability assessors guidelines for the consistent classification of failure for assessing the Reliability requirements of the Cannon, 155MM: M199. This Appendix B is a mandatory part of the specifciation. The information contained herein is intended for compliance.

20. APPLICABLE DOCUMENTS

STANDARDS

MILITARY

MIL-STD-721 - Definitions of Effectiveness Terms for Reliablitity, Maintainability, Human Factors, and Safety

MIL-STD-882 - System Safety Program Requirements

30. REQUIREMENTS

30.1 Reliability. The cannon as a component of a fully operational Howitzer, 155MM: M198 Howitzer system, shall have for the Operational Mode Summary/Mission Profile (OMS/MP) (see 3.3) a minimum acceptable value (MAV) of 2,100 Equivalent Full Charge (EFC) mean rounds between failure (MRBF) with 80 percent confidence (see 40.2). The scoring criteria for assessing the reliability requirements new stated in 40.6.

30.1.1 Mission profile. A typical 48 hour mission consists of firing 250 mixed charge rounds which are equivalent to 100 EFC rounds.

30.2 Maintainability. The cannon parts and components shall have the following maintainabilities:

a. Tube: Cleaning and Lubricatin the tube after firing, in accordance with applicable maintenance procedures, to prevent corrosion shall be the only maintenance required.

b. Breech: Service of the breech shall be carried out by the crew and shall be limited to minor maintenance and cleanup. Disassembly, cleaning and reassembly of the breechlock, firing mechanism block assembly, and associated operating linkage shall be carried out by the crew within 1.5 man-hours. Repair shall be carried out at unit (company or battalion) level.

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30.3 Durability. The cannon parts and components shall have the following durability (see 50.1 and 50.2):

a. Breech ring and breechblock - 15,000 EFC rounds.

b. Tube fatigue life - 10,000 EFC rounds.

c. Tube wear life - For tactical purposes a tube is worn out when the diameter of the bore measured at 41.75 inches from the rear face of the tube increases by .100 inches or when the diameter of the bore measured at 60 inches from the rear face of this tube increases by .060 inches. This wear limit occurs at a mean life of 1750 rounds when firing the M203 propelling charge at 70°F.

d. Muzzle brake life - Equal to the tube wear life (see 30.3.c).

e. All other parts and components will be replaced as required in the scheduled maintenance.

40. INSPECTION METHODS

40.1 Pre-reliability test inspections. Upon completion of climatic and thermal warning device tests (see Appendix A), all examinations (see 4.7.4) and functional inspections (see 4.7.5) shall be repeated before reliability and durbility tests are performed.

40.2 Reliability test. The test shall be conducted on two cannon, Each cannon shall be fired 1690 times for a total of 3380 rounds. Ammunition shall be inert loaded M483A1 projectiles with dummy PD M51A5 fuzes, M82 primers, and zone 8 M203 charge, at 70°F ± 3°F. To demonstrate that the reliability requirements have been achieved with 80 percent confidence, this test shall be completed without failure. Failure scoring criteria for assessing reliability requirements shall be in accordance with 40.6. If the requirements of 30.1 are not complied with, acceptance shall be deferred and paragraph 40.4 applies.

NOTE ; Rounds fired in other tests (see Appendix A), may be counted as rounds for this test.

40.3 Durability test. The durability test shall be conducted on two cannon. Ammunition shall be inert loaded M483A1 projectiles with dummy PD M51A5 fuzes, M82 primers, and zone 8 M203 charge, at 70°F±3°F. Rounds fired in the reliability test (see 40.2) may be counted towards the rounds necessary for this test. If the requirements of 30.3 are not complied with, acceptance shall be deferred and paragraph 40.4 applies.

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40.3.1 Tube. Durability shall be demonstrated by measureing wear after every 250 rounds until 0.100 inch wear has occurred at 41.75 inches from the rear face of the tube, or until 0.060 inch wear has occurred at 60 inches from the rear face of the tube. The firing rate shall not exceed one round every two minutes and shall be such that the thermal warning device does not exceed 250°F. If the preceding wear occurs in either *or* both tubes before 1750 rounds has been reached, paragraph 40.5 applies.

40.3.2 Muzzle brake. After firing 1750 rounds on each cannon, the muzzle brake shall be magnetic particle inspected in accordance with applicable drawings and specifications.

40.3.3 Breech mechanism. Durability shall be demonstrated by dynamic presure cycling the breech mechanism at Watervliet Arsenal or other approved facility, a total of 15,000 EFC pressure cycles. Ronnds fired in other tests of 40.1 and 40.2 shall be counted as cycles for this test.

40.4 Failure analysis. The contractor shall conduct a failure analysis by performing a dimensional, physical and visual examination of the components which are suspected to have caused the failure. Results shall be submitted to the Government for a failure cause determination. If it is determined that failure occurred because of deviation from specified dimensions *or* physical properties or both, the contractor shall accomplish, as applicable, the following action:

a. Evaluate and correct the applicable production processes and procedures to prevent recurrence of the same defect in future production.

b. Examine completely assembled cannon and partially assembled cannon and components to ensure that material containing the same defect is purged from the inventory and not presented to the Government for acceptance.

c. Submit the results of the failure analysis and the corrective actions taken to the Government for evaluation and approval of retest or continuation of test.

40.5 Tube Wear Data. If tube requirements are not met, (see 40.3.1) the results shall analyzed in accordance with currently available wear data for the M199 tube and a determination made as to whether the results indicate performance substantially below expected levels. If it is determined the results so indicate, then the following procedures shall apply:

a. Conduct a failure analysis study performing a dimensional, visual and metallurgical examination of the tube.

b. Review the ammunition elements (propellant, charge, and projectile characteristics) for abnormalities or changes in design or configuration since last tested.

c. Evaluate applicable production processes and procedures and correct as warranted.

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d. Examine tubes at in-process and final assembly point to assure that defects are purged from the inventory and are not presented to the Government for acceptance.

e. Submit results of the failure analysis and corrective actions taken to the Government for evaluation and approval to retest or continue the test.

40.6 Scoring Criteria. The failure scoricng procedure (see 40.6.5) provides for the use of the failure definition (see 50.4) which is amplified (see 40.6.1). Failure criteria (see 40.6.2), degraded modes of operation (see 40.6.3) and failure exclusions (see 40.6.4) pertaining to the cannon provide additional failure scoring guidelines.

40.6.1 Failure definition amplification.

40.6.1.1 The following are not considered as reliability failures:

a. Scheduled replacement of parts before failure.

b. An incipient-malfunction corrected during scheduled preventive maintenance on the part in question provided a higher level of maintenance is not necessary.

c. A malfunction resulting from not following the prescribed operational-or maintenance procedures or schedule dictated by the equipment manuals.

d. A malfunction resulting from test item abuse, unrealistic operation conditions or accident;

e. Actual or incipient malfunctions detected or corrected during initial technical inspections and incipient malfunction detected during final technical inspection.

40.6.1.2 The following are considered as reliability failures:

a. A failure detected and/or corrected during the correction of another failure provided the failures are totally unrelated.

b. Corrected incipient malfunctions not covered by 40.6.1.1 examples 2 and 5.

c. Failures resulting from lack of clarity of instruction or other fault in the maintenance test package.

40.6.2 Failure Modes. The following are two typical failure modes for use in assessment of mission reliability failures. These failure modes are subject to performance degradation (see 40.6.3).

a. Breechblock cannot be closed/opened.

b. Gun will not fire.

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40.6.3 Degraded modes of operation. A certain amount of performance degradation must take place before a reliability failure can be assumed, If the system ceases to function, performance becomes unacceptable (i.e., non-existent); however, if performance-is only degraded, then some threshold must be extablished to define where reliability failures begin. Allowable degradation thresholds have been established for the various performance parameters. Performance beyond these thresholds constitutes a reliability failure. As long as performance remains above the threshold or is returned above the threshold within five minute maintenance period, no reliability failure is charged.

a.	Rate of Fire - maximum	3 rounds/rein
b.	Rate of Fire - sustained	1 round/1.5 min

40.6.4 Failure exclusions. Based on 14 November 1973, Scoring Conferences: Training and Doctrine Command (TRADOC) representatives did not wish to have obturator pad and split rings replaced on a scheduled basis. The TRADOC preferred approach is to leave parts in until no longer serviceable to get maximum service from each part, but not count these part replacements as reliability failures. TRADOC wants expected obturator life data in TM's to supplement the explanation of conditions requiring replacement, however, there should not be a prescribed replacement schedule in the TM's. ARRCOM concurs.

40.6.5 Failure scoring procedure. The following scheme provides the instructions for the use of the failure definition (see 50.4) for assessing reliability requirements of the Cannon, 155MM Howitzer: M199 (see 30.1). These instructions assume that an incident or malfunction has occurred during test, and a record has been prepared against this action to the extent where it is necessary to determine the chargeability of the action. Incidents where the identification of a reliability failure is in doubt will be deferred to a user/developer/tester Scoring Conference. The step-by-step instructions for using the failure definition are as follows:

Step 1. Board the incidnet information whenn a malfunction of the cannon occurs or is detected (rounds on test item and system, description of incident, nature of maintenance action required, maintenance time in hours, maintenance manhours, etc.). proceed to the next step answeing the question posted and takng action according to the answer given.

Step 2. Is this a scheduled replacement of parts before failure? If yes, do not charge a failure; if no, proceed to Step 3.

Step 3. Is this a malfunction resulting from not following the prescribed operational or maintenance procedures or schedule dictated by the equipment manuals? If yes, do not charge a failure; if no, proceed to Step 4.

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Step 4. Is this malfunction, resulting from test item abuse, unrealistic operating conditions or accident? If yes, do not charge a failure: if no, proceed to Step 5.

Step 5. Is this an actual or incipient malfunction detected or corrrected during initial technical inspection or an incipient malfunction detected during final technical inspection? If yes, do not charge a failure; if no, proceed to Step 6.

Step 6. Is this an incipient malfunction corrected during scheduled preventive maintenance on the part in question, provided a higher level of maintenance is not necessary? If yes, do not charge a failure; if no, proceed to Step 7.

Step 7. Is this incident related to or caused by another reliability failure? If yes, do not charge a failure; if no, proceed to Step 8.

Step 8. The following statements are addressed collectively, (A) The malfunction did not and would not cause a critical or catastrophic hazard to personnel or equipment as defined by MIL-STD-882. (B) The system performance is above acceptable levels (see 40.6.3). (C) The malfunction did not and would not preclude the ability to commence or cease any mode of operation (e.g., load and fire at any range, charge, elevation, or direction, (see 40.6.2). If all three all three of the above statements are true, do not charge a reliability failure; if any are not true, proceed to Step 9.

Step 9. Is the operator/crew authorized and able to rememdy the malfunction by adjustment, repair, or replacement action within five minutes. using the controls. Basic Issue Items (611), Items Troop Installed or Authorized (ITIA) and parts authorized to the crew? If the answer is yes, do not charge a reliability failure; if no, proceed to Step 10.

Step 10. If the previous 9 steps provide inconclusive evidence as to the chargeability or non-chargeability of a reliability failure, the incident should be checked against the criteria of 40.6.1 - 40.6.4. If categorization of the incident remains in doubt, defer classification to a user/developer/tester Scoring Conference.

50. Definitions.

50.1 RAM-D terms and definitions. Reliability, availability, maintaninability and durability terms and definitions-used in this specification are in accordance with MIL-STD-721. Durability is further defined in 50.2

50.2 Durability. Durability is the probility that an item will successfully survive Its projected life, overhaul point, or rebuild point (whichever is the more appropriate durability measure for the item) without a durability failure. A durability failure is considered to be a malfunction that precludes further operation of the item and is great enough in cost, safety, or time to restore, that the item must be replaced or rebuilt.

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APPENDIX B

50.3 EFC (Equivalent Full Charge). An EFC round is a zone 8 M203 charge, utilizing a M483A1 projectile and fired at 70″F ± 3°F.

50.4 Failure definition. For the purpose of assesing reliability, a failure is defined as any malfunction which the operator or crew cannot or is not authorized to remedy by adjustment, repair, or replacement action within five minutes, using the controls, Basis Issue Items (BII), Items Troop Installed or Authorized (ITIA) and parts authorized to the crew, and which causes or would cause: 1. inability to commence or cease any mode of operation, 2. a critical or catastrophic hazard to personnel or equipment as defined by MIL-STD-882. For related malfunctions only the primary malfunction will be counted against reliability (see 40.6.1).

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-45975B	CANNON, 155MM HOWITZER: M199

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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